U.S.N.										

B.M.S. College of Engineering, Bengaluru-560019

Autonomous Institute Affiliated to VTU

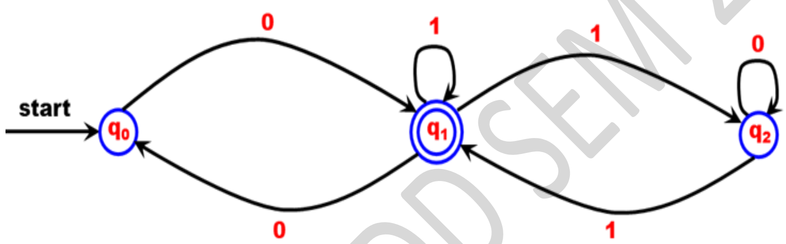
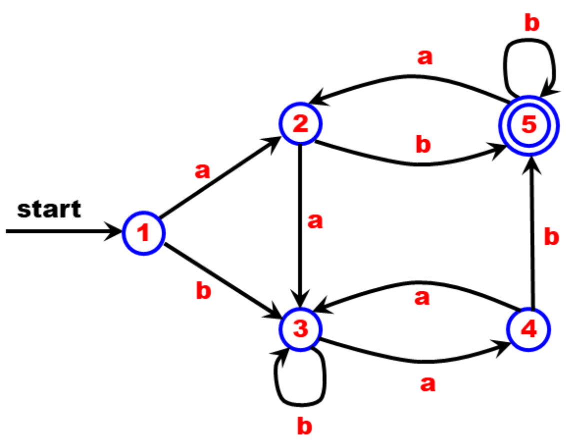
April 2024 Semester End Main Examinations

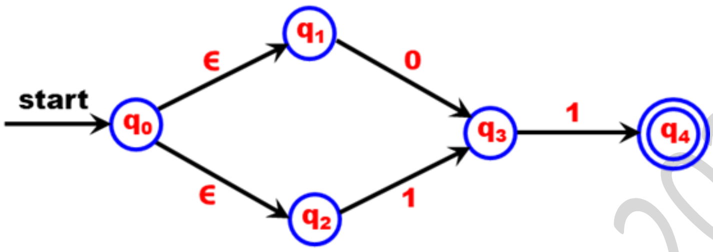
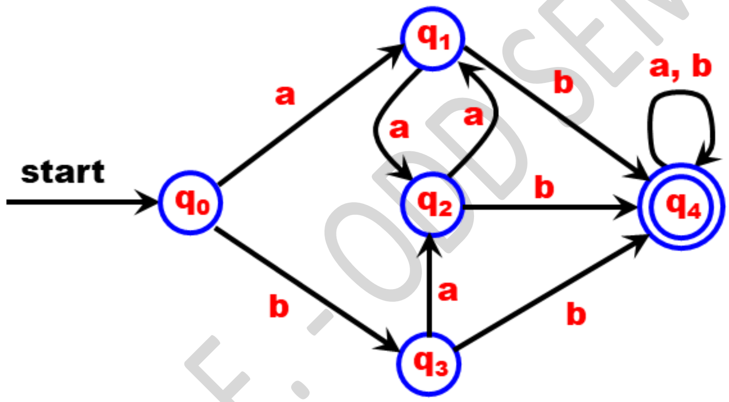
Programme: B.E.
Branch: Information Science and Engineering
Course Code: 22IS4PCTFC
Course: Theoretical Foundations of Computation

Semester: IV
Duration: 3 hrs.
Max Marks: 100

Instructions: 1. Answer any FIVE full questions, choosing one full question from each unit.
2. Missing data, if any, may be suitably assumed.

Important Note: Completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages. Revealing of identification, appeal to evaluator will be treated as malpractice.

		UNIT - I	***CO***	***PO***	**Marks**
1	a)	Convert the following NFA to its equivalent DFA: 	CO1	PO1	**07**
	b)	Design DFA's for the following Languages: (i) Design a DFA to accept all the strings containing the substring 011 when $\Sigma = \{0, 1\}$ (ii) Design a DFA to accept all the strings in which the leftmost symbol differs from the rightmost symbol when $\Sigma = \{0, 1\}$	CO3	PO3	**06**
	c)	Minimize the following DFA: 	CO1	PO1	**07**

<table>
<tr><td></td><td></td><td></td><td>OR</td><td></td><td></td><td></td></tr>
<tr><td></td><td>2</td><td>a)</td><td>Differentiate NFA and DFA. Design DFA's for the following Languages:
(i) Design a DFA to accept all the strings ending with 1 and does not contain the substring 00 when $\Sigma = \{0, 1\}$
(ii) Design a DFA to accept strings of 0's and 1's starting with at least two 0's and ending with at least two 1's when $\Sigma = \{0, 1\}$</td><td>CO3</td><td>PO3</td><td>06</td></tr>
<tr><td></td><td></td><td>b)</td><td>Convert the following ε-NFA to its equivalent DFA:

</td><td>CO1</td><td>PO1</td><td>07</td></tr>
<tr><td></td><td></td><td>c)</td><td>Minimize the following DFA:

</td><td>CO1</td><td>PO1</td><td>07</td></tr>
<tr><td></td><td></td><td></td><td>UNIT - II</td><td></td><td></td><td></td></tr>
<tr><td></td><td>3</td><td>a)</td><td>(i) Obtain a Regular Expression to accept all the strings where the third last symbol is 1. Consider $\Sigma = \{0, 1\}$
(ii) Obtain a Regular Expression to accept all the strings having at least one double 0 or double 1. Consider $\Sigma = \{0, 1\}$</td><td>CO2</td><td>PO2</td><td>06</td></tr>
<tr><td></td><td></td><td>b)</td><td>State and Prove Pumping Lemma Theorem for regular languages. Also, show that the Language
$L = \{ww^R \mid w \in (0 + 1)^* \}$ is not regular.</td><td>CO2</td><td>PO2</td><td>07</td></tr>
<tr><td></td><td></td><td>c)</td><td>Design ε-NFA corresponding to the Regular Expression
i) $((0 + 1)^* + 01)^*$
ii) $01+1^*0$</td><td>CO1</td><td>PO1</td><td>07</td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
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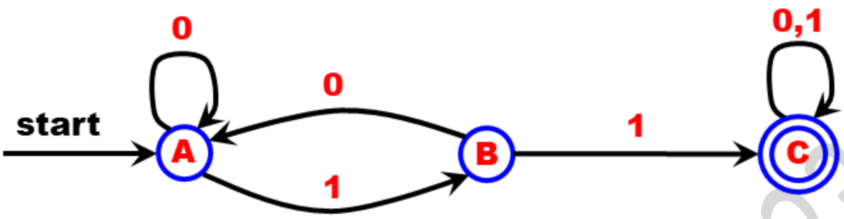
			OR			
	4	a)	(i) Obtain a Regular Expression to accept all the strings that either start with 01 or end with 01. Consider Σ = {a, b} (ii) Obtain a Regular Expression to accept all the strings starting with a, but not having consecutive b's. Consider Σ = {0, 1}	CO2	PO2	**06**
		b)	Show that the Language L = { $a^i b^j$ | i > j } is not Regular.	CO2	PO2	**07**
		c)	Obtain the Regular Expression for the given Finite Automata using State Elimination Method:  	CO2	PO2	**07**
			UNIT - III			
	5	a)	Design CFG's to generate the following Languages: (i) L = { w | w containing 00 or 11 when Σ = {0, 1} } (ii) L = { $a^m b^n$, n > m, n >= 1, m >= 1 }	CO3	PO3	**06**
		b)	Show that the following Grammar is Ambiguous: S → SbS | Sa | bSS | SSb | a	CO2	PO2	**06**
		c)	Simplify the following Grammar: G = (V, T, P, S) V = { S, A, B } T = { a } P = { S → AAA | B A → aA | B B → ε} S = { S }	CO1	PO1	**08**
			UNIT - IV			
	6	a)	Differentiate between Deterministic PDA and Non- deterministic PDA.	CO1	PO1	**05**
		b)	Design a PDA to accept the Language: L = { w | w ε (a + b)* where number of a's in w > number of b's in w }	CO3	PO3	**08**
		c)	Convert the following CFG to the equivalent PDA: S → aABD A → aB | b B → bA | a C → a	CO1	PO1	**07**

			UNIT - V			
	7	a)	Design a Turing Machine to accept a given string w and generates the string ww where w ϵ a^+ by final state acceptance.	CO3	PO3	**08**
		b)	Design a Turing Machine to accept the Language L = { 0^n1^n | n >= 1 } by final state acceptance.	CO3	PO3	**07**
		c)	Discuss how Turing Machines can be combined to perform complicated tasks with an example.	CO1	PO1	**05**

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